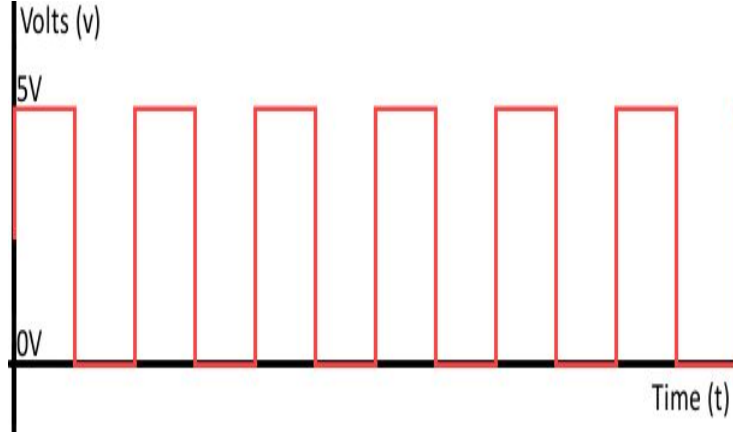


DIGITAL DESIGN 1

- Digital Logic and Computer Design by M Moris Mano
- Fundamentals of Digital Circuits by Anand Kumar
- Modern Digital Electronics by R P Jain

DIGITAL SIGNAL



DISCRETE SIGNAL

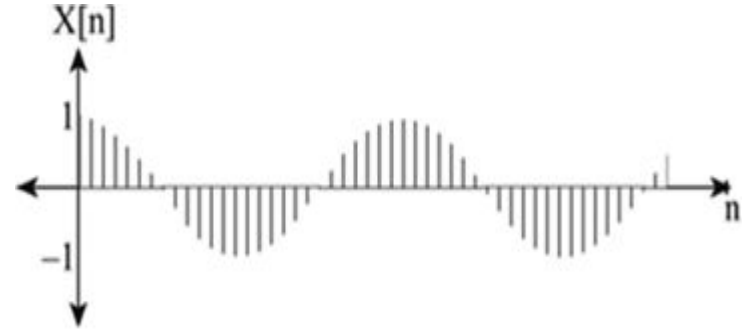
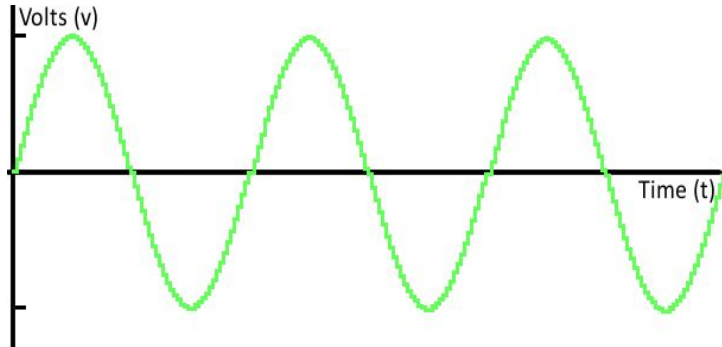


Figure 1

ANALOG SIGNAL



Amplitude: Analog and Digital Signal
Time: Continuous and Discrete Signal

WHY DIGITAL SYSTEMS?

- Easier to design
- Easy Information storage
- Greater Accuracy and Precision
- More versatile
- Less affected by noise
- More digital circuitry can be fabricated on IC chips

★ **But real world is Analog**

NUMBER SYSTEM

Numbering Systems		
System	Base	Digits
Binary	2	0 1
Octal	8	0 1 2 3 4 5 6 7
Decimal	10	0 1 2 3 4 5 6 7 8 9
Hexadecimal	16	0 1 2 3 4 5 6 7 8 9 A B C D E F

Base or radix – r

$$d_i = \{0, 1, \dots, r - 1\}$$

$$d_n \dots d_1 d_0 . d_{-1} d_{-2} \dots d_{-m}$$

$$= d_n \times r^n + \dots + d_1 \times r^1 + d_0 \times r^0$$

$$+ d_{-1} \times r^{-1} + d_{-2} \times r^{-2} + \dots + d_{-m} \times r^{-m}$$

DECIMAL NUMBER SYSTEM

- Positional weighted system

$$d_n \ d_{n-1} \ d_{n-2} \ \dots \ d_1 \ d_0 \ . \ d_{-1} \ d_{-2} \ d_{-3} \ \dots \ d_{-k}$$

$$(d_n \times 10^n) + (d_{n-1} \times 10^{n-1}) + \dots + (d_1 \times 10^1) + (d_0 \times 10^0) + (d_{-1} \times 10^{-1}) \\ + (d_{-2} \times 10^{-2}) + \dots$$

$$2586_{10} = (2 \times 10^3) + (5 \times 10^2) + (8 \times 10^1) + (6 \times 10^0)$$

$$= 2000 + 500 + 80 + 6$$

BINARY NUMBER SYSTEM

- Positional Weighted System
- A **binary digit** is called a **bit**

$d_n \ d_{n-1} \ d_{n-2} \ \dots \ d_1 \ d_0 \ . \ d_{-1} \ d_{-2} \ d_{-3} \ \dots \ d_{-k}$

Its decimal equivalent is

$$(d_n \times 2^n) + (d_{n-1} \times 2^{n-1}) + \dots + (d_1 \times 2^1) + (d_0 \times 2^0) \\ + (d_{-1} \times 2^{-1}) + (d_{-2} \times 2^{-2}) + \dots$$

$$\begin{aligned} 10101_2 &= (1 \times 2^4) + (0 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0) \\ &= 16 + 0 + 4 + 0 + 1 \\ &= 21_{10} \end{aligned}$$

OCTAL NUMBER SYSTEM

- Positional Weighted System

The decimal value of the octal number $d_n d_{n-1} d_{n-2} \dots d_1 d_0 \cdot d_{-1} d_{-2} \dots d_{-k}$ is
 $(d_n \times 8^n) + (d_{n-1} \times 8^{n-1}) + \dots + (d_1 \times 8^1) + (d_0 \times 8^0) + (d_{-1} \times 8^{-1}) + (d_{-2} \times 8^{-2}) + \dots$

$$\begin{aligned} 2057_8 &= (2 \times 8^3) + (0 \times 8^2) + (5 \times 8^1) + (7 \times 8^0) \\ &= 1024 + 0 + 40 + 7 \\ &= 1071_{10} \end{aligned}$$

HEXADECIMAL NUMBER SYSTEM (Hex)

- Positional Weighted System
- Can represent binary number concisely

If the hex number is $d_n d_{n-1} \dots d_1 d_0 . d_{-1} d_{-2} \dots d_{-k}$, its decimal equivalent is $(d_n \times 16^n) + (d_{n-1} \times 16^{n-1}) + \dots + (d_1 \times 16^1) + (d_0 \times 16^0) + (d_{-1} \times 16^{-1}) + (d_{-2} \times 16^{-2}) + \dots$

$$\begin{aligned} 1AF_{16} &= (1 \times 16^2) + (A \times 16^1) + (F \times 16^0) \\ &= 1 \times 256 + 10 \times 16 + 15 \times 1 \\ &= 256 + 160 + 15 \\ &= 431_{10} \end{aligned}$$

CONVERSION OF A NUMBER OF ANY BASE TO DECIMAL NUMBER

Step 1: Determine the column (positional) value of each digit

Step 2: Multiply the obtained column values by the digits in the corresponding columns

Step 3: Calculate the sum of these products

EXAMPLE

$$4706_8 = ?_{10}$$


$$\begin{aligned} 4706_8 &= 4 \times 8^3 + 7 \times 8^2 + 0 \times 8^1 + 6 \times 8^0 \\ &= 4 \times 512 + 7 \times 64 + 0 + 6 \times 1 \\ &= 2048 + 448 + 0 + 6 \leftarrow \text{Sum of these products} \\ &= 2502_{10} \end{aligned}$$

Common
values
multiplied
by the
corresponding
digits

$$2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad 2^0 \quad 2^{-1} \quad 2^{-2} \quad 2^{-3}$$

$$\begin{aligned} 1 \quad 1 \quad 0 \quad 1 \quad 1 \cdot 1 \quad 0 \quad 1 &= (1 \times 2^4) + (1 \times 2^3) + (0 \times 2^2) + (1 \times 2^1) + (1 \times 2^0) \\ &\quad + (1 \times 2^{-1}) + (0 \times 2^{-2}) + (1 \times 2^{-3}) \\ &= 16 + 8 + 0 + 2 + 1 + 0.5 + 0 + 0.125 \\ &= 27.625_{10} \end{aligned}$$

CONVERSION OF A DECIMAL NUMBER TO NUMBER OF ANY BASE

Division-Remainder Method

Step 1: Divide the decimal number to be converted by the value of the new base

Step 2: Record the remainder from Step 1 as the rightmost digit (least significant digit) of the new base number

Step 3: Divide the quotient of the previous divide by the new base

Step 4: Record the remainder from Step 3 as the next digit (to the left) of the new base number

Repeat Steps 3 and 4, recording remainders from right to left, until the quotient becomes zero in Step 3

Note that the last remainder thus obtained will be the most significant digit (MSD) of the new base number

EXAMPLE

EXAMPLE 2.11 Convert 105.15_{10} to binary.

Solution

Conversion of integer 105.

Successive division

Remainder

2	105	
2	52	1
2	26	0
2	13	0
2	6	1
2	3	0
2	1	1
	0	1

Reading the remainders from bottom to top, $105_{10} = 1101001_2$.

Conversion of fraction 0.15_{10}

Given fraction

			0.15
Multiply	0.15	by 2	0.30
Multiply	0.30	by 2	0.60
Multiply	0.60	by 2	1.20
Multiply	0.20	by 2	0.40
Multiply	0.40	by 2	0.80
Multiply	0.80	by 2	1.60

EXAMPLE 2.44 Convert 378.93_{10} to octal.

Solution

Conversion of 378_{10} to octal

Successive division	Remainders
$\begin{array}{r l} 8 & 378 \\ 8 & 47 \\ 8 & 5 \\ & 0 \end{array}$	$\begin{array}{c} \uparrow \\ 2 \\ \\ 7 \\ \\ 5 \end{array}$

Read the remainders from bottom to top. Therefore, $378_{10} = 572_8$

Conversion of 0.93_{10} to octal

0.93×8	$\downarrow$	7.44
0.44×8		3.52
0.52×8	$\downarrow$	4.16
0.16×8		1.28

Read the integers to the left of the octal point downwards.
Therefore, $0.93_{10} = 0.7341_8$. Hence $378.93_{10} = 572.7341_8$.

EXAMPLE 2.60 Convert 2598.675_{10} to hex.

Solution

The given decimal number is a mixed number. Convert the integer and the fraction part separately to hex.

Conversion of 2598_{10}

Successive division

16		2598
16		162
16		10
		0

Remainder	
Decimal	Hex

6	↑	6
2		2
10		A

Reading the remainders upwards, $2598_{10} = A26_{16}$.

Conversion of 0.675_{10}

Given fraction is 0.675

0.675×16		10.8
0.800×16		12.8
0.800×16		12.8
0.800×16	↓	12.8

Reading the integers to the left of hexadecimal point downwards, $0.675_{10} = 0.ACCC_{16}$.
Therefore, $2598.675_{10} = A26.ACCC_{16}$.

CONVERSION OF A NUMBER FROM ONE BASE TO ANOTHER BASE

Step 1: Convert the original number to a decimal number (base 10)

Step 2: Convert the decimal number so obtained to the new base number

EXAMPLE

$$545_6 = ?_4$$

Solution:

Step 1: Convert from base 6 to base 10

$$\begin{aligned} 545_6 &= 5 \times 6^2 + 4 \times 6^1 + 5 \times 6^0 \\ &= 5 \times 36 + 4 \times 6 + 5 \times 1 \\ &= 180 + 24 + 5 \\ &= 209_{10} \end{aligned}$$

Step 2: Convert 209_{10} to base 4

4	209	Remainders
	52	1
	13	0
	3	1
	0	3

Hence, $209_{10} = 3101_4$

So, $545_6 = 209_{10} = 3101_4$

Thus, $545_6 = 3101_4$

BINARY TO OCTAL CONVERSION

Step 1: Divide the digits into groups of three starting from the right

Step 2: Convert each group of three binary digits to one octal digit using the method of binary to decimal conversion

EXAMPLE

$$1101010_2 = ?_8$$

Step 1: Divide the binary digits into groups of 3 starting from right

$$\underline{001} \quad \underline{101} \quad \underline{010}$$

Step 2: Convert each group into one octal digit

$$001_2 = 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 1$$

$$101_2 = 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 5$$

$$010_2 = 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 = 2$$

$$\text{Hence, } 1101010_2 = 152_8$$

OCTAL TO BINARY CONVERSION

Step 1: Convert each octal digit to a 3 digit binary number (the octal digits may be treated as decimal for this conversion)

Step 2: Combine all the resulting binary groups (of 3 digits each) into a single binary number

EXAMPLE

$$562_8 = ?_2$$

Step 1: Convert each octal digit to 3 binary digits

$$5_8 = 101_2, \quad 6_8 = 110_2, \quad 2_8 = 010_2$$

Step 2: Combine the binary groups

$$562_8 = \begin{array}{ccc} \underline{101} & \underline{110} & \underline{010} \\ 5 & 6 & 2 \end{array}$$

$$\text{Hence, } 562_8 = 101110010_2$$

BINARY TO HEXADECIMAL CONVERSION

Step 1: Divide the binary digits into groups of four starting from the right

Step 2: Combine each group of four binary digits to one hexadecimal digit

EXAMPLE

$$111101_2 = ?_{16}$$

Step 1: Divide the binary digits into groups of four starting from the right

0011

1101

Step 2: Convert each group into a hexadecimal digit

$$0011_2 = 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 = 3_{10} = 3_{16}$$

$$1101_2 = 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 13_{10} = D_{16}$$

Hence, $111101_2 = 3D_{16}$

HEXADECIMAL TO BINARY CONVERSION

Step 1: Convert the decimal equivalent of each hexadecimal digit to a 4 digit binary number

Step 2: Combine all the resulting binary groups (of 4 digits each) in a single binary number

EXAMPLE

$$2AB_{16} = ?_2$$

Step 1: Convert each hexadecimal digit to a 4 digit binary number

$$2_{16} = 2_{10} = 0010_2$$

$$A_{16} = 10_{10} = 1010_2$$

$$B_{16} = 11_{10} = 1011_2$$

Step 2: Combine the binary groups

$$2AB_{16} = \begin{array}{ccc} \underline{0010} & \underline{1010} & \underline{1011} \\ 2 & A & B \end{array}$$

$$\text{Hence, } 2AB_{16} = 001010101011_2$$